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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2017 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 2

Wednesday 22nd February 2017

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eight.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 74 boys

Examiner

LRP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Given that $z = x + iy$, which expression is equal to $|z|$?

- (A) $x + y$
- (B) $z\bar{z}$
- (C) $x^2 + y^2$
- (D) $\sqrt{z\bar{z}}$

QUESTION TWO

Given that $z = i\sqrt{3} - 1$, which of the following represents the complex number z ?

- (A) $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
- (B) $2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
- (C) $2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
- (D) $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$

QUESTION THREE

Which expression is equal to $\int \frac{1}{\sqrt{6x - x^2}} dx$?

- (A) $\sin^{-1}\left(\frac{x+3}{3}\right) + C$
- (B) $-\sin^{-1}\left(\frac{x+3}{3}\right) + C$
- (C) $\sin^{-1}\left(\frac{x-3}{3}\right) + C$
- (D) $\sin^{-1}\left(\frac{3-x}{3}\right) + C$

QUESTION FOUR

Which of the following is a primitive of $\cos^3 x \sin x$?

(A) $4 \cos^4 x - 3 \cos^2 x$

(B) $\frac{\cos^4 x}{4}$

(C) $\frac{\sin^4 x}{4} - \frac{\sin^2 x}{2}$

(D) $-\frac{\cos^4 x}{4}$

QUESTION FIVE

Given $x^2y - y^2 = 1$, which of the following is the correct result for $\frac{dy}{dx}$?

(A) $\frac{1 - 2xy}{x^2 - 2y}$

(B) $\frac{2xy}{x^2 + 2y}$

(C) $\frac{2xy}{x^2 - 2y}$

(D) $\frac{2xy}{2y - x^2}$

QUESTION SIX

Which expression is equal to $\int \frac{dx}{e^x + 1}$?

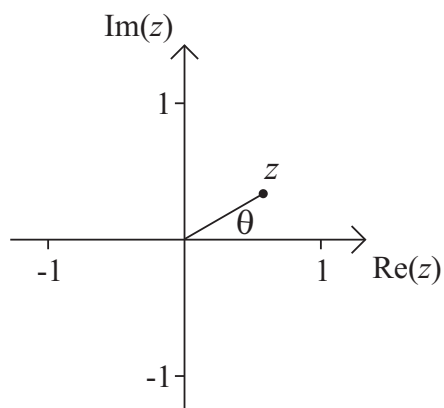
(A) $-\frac{e^x}{(e^x + 1)^2} + C$

(B) $x - \log_e(e^x + 1) + C$

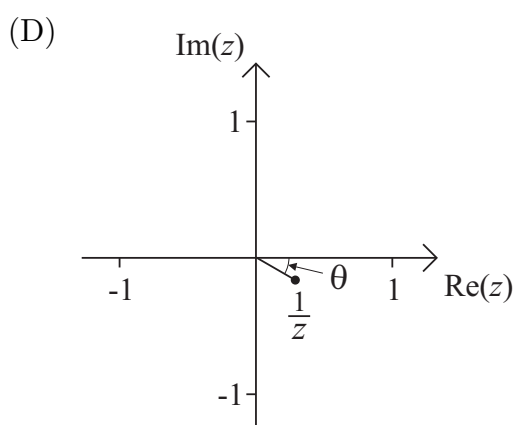
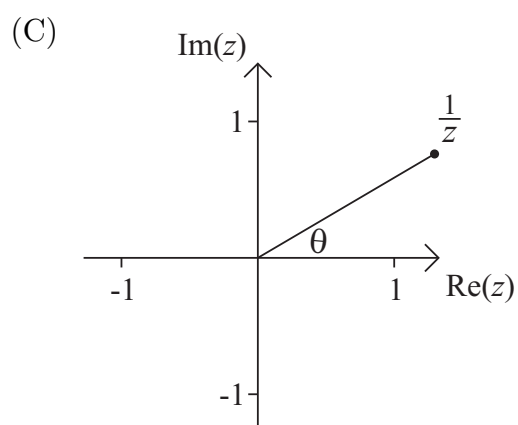
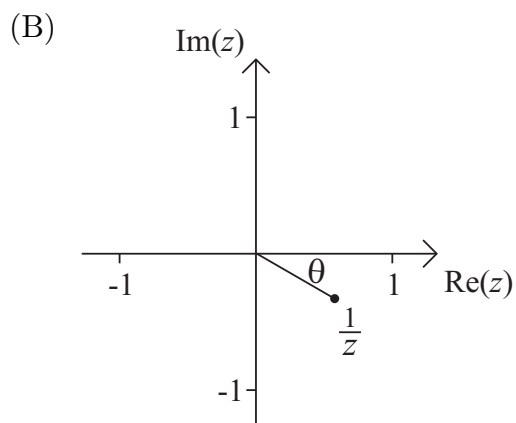
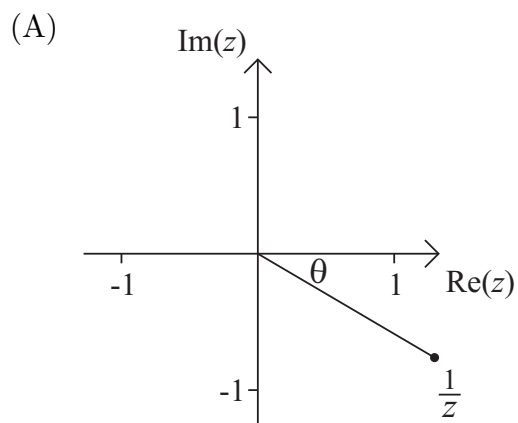
(C) $x - \frac{1}{e^x} + C$

(D) $\frac{\log_e(e^x + 1)}{e^x} + C$

QUESTION SEVEN



The Argand diagram above shows the complex number z . Which of the following diagrams best shows the complex number $\frac{1}{z}$?



_____ End of Section I _____

Examination continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet.	Marks
(a) Let $w = -1 + i$ and $z = 4 + 3i$.	
(i) Find $2w - \bar{z}$.	1
(ii) Find $\text{Im}\left(\frac{z}{i}\right)$.	1
(iii) Express $\frac{w}{z}$ in the form $a + ib$, where a and b are real numbers.	2
(b) (i) Find the two square roots of $5 - 12i$.	2
(ii) Hence, or otherwise, solve $z^2 - (1 - 4i)z - (5 - i) = 0$.	2
(c) Let $z = \cos \theta + i \sin \theta$, where $\frac{\pi}{4} < \theta < \frac{\pi}{2}$. Plot points representing the positions of z , $2z$ and z^2 on an Argand diagram.	2
(d) Sketch the locus defined by the intersection of $0 \leq \arg(z + 2) \leq \frac{\pi}{6}$ and $\text{Re}(z) \geq 0$.	2

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) (i) Use the compound-angle formulae to show that

1

$$\cos \alpha \sin \beta = \frac{\sin (\alpha + \beta) - \sin (\alpha - \beta)}{2}.$$

- (ii) Hence evaluate $\int_0^{\pi} \cos 3x \sin 2x \, dx$.

2

- (b) Find:

(i) $\int \frac{6x^3 + 3x^2 - 2}{2x + 1} \, dx$

2

(ii) $\int \sin^3 x \cos^2 x \, dx$.

2

(c) Evaluate $\int_1^e \frac{(\log_e x)^2}{x} \, dx$.

2

(d) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{3}} \frac{1}{2 - \cos x} \, dx$.

3

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) (i) Find real numbers A , B , C and D such that

2

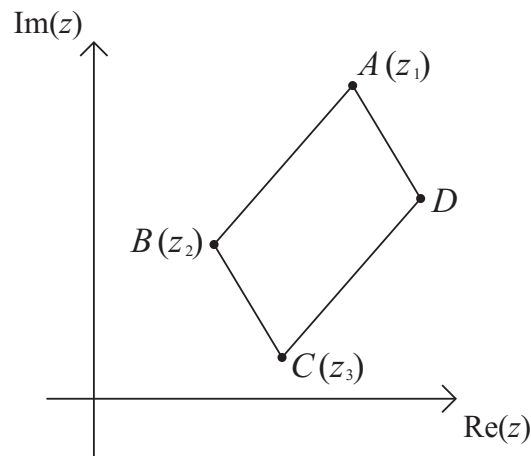
$$\frac{x}{x^4 - 1} \equiv \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{Cx + D}{x^2 + 1}.$$

- (ii) Hence find $\int \frac{x}{x^4 - 1} dx$.

2

- (b)

1



In the Argand diagram above, $ABCD$ is a parallelogram. The points A , B and C represent the complex numbers z_1 , z_2 and z_3 respectively. Find the complex number represented by the vector $\overrightarrow{BD}$ in terms of z_1 , z_2 and z_3 .

- (c) A complex number z satisfies $|z + 4 - 3i| = 2$.

- (i) Sketch the locus of z in the complex plane.

1

- (ii) Find the maximum value of $|z|$.

1

- (iii) Determine the values of k for which the simultaneous equations

1

$$|z + 4 - 3i| = 2 \quad \text{and} \quad |z + 4| = k$$

have exactly one solution for z .

- (d) (i) Use a suitable substitution to show that $\int e^{\sqrt{x}} dx = \int 2ue^u du$.

1

- (ii) Hence use integration by parts to evaluate $\int_0^4 e^{\sqrt{x}} dx$.

3

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

(a) A complex number z satisfies $|z - 2| = 2$ and $0 < \arg(z) < \frac{\pi}{2}$.

(i) Use a diagram to show that $\arg(z - 2) = 2 \arg(z)$.

1

(ii) Show that $\arg(z^2 - 2z) = \frac{3 \arg(z - 2)}{2}$.

1

(b) Let $f(x) = \tan^{-1}\left(x + \frac{1}{x}\right)$, where $x > 0$.

(i) Find the coordinates of the minimum stationary point. You do not have to justify that it is a minimum turning point.

2

(ii) Find the equation of the horizontal asymptote.

1

(iii) Find $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^+} f'(x)$.

2

(iv) Sketch the graph of the function.

1

(c) Let $z = x + iy$ be a complex number, where x and y are real numbers. Suppose that z satisfies

$$z\bar{z} + (1 - 2i)z + (1 + 2i)\bar{z} \leq 3.$$

(i) Find the equation of the locus of z and sketch this locus on an Argand diagram.

2

(ii) Find the maximum and minimum values of $x + y$.

2

HINT: Consider the line $x + y = k$.

————— End of Section II —————

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

EXTENSION 2

Half Yearly 2017 - SOLUTIONS

Q1. D

Q2. C

Q3. C

Q4. D

Q5. D

Q6. B

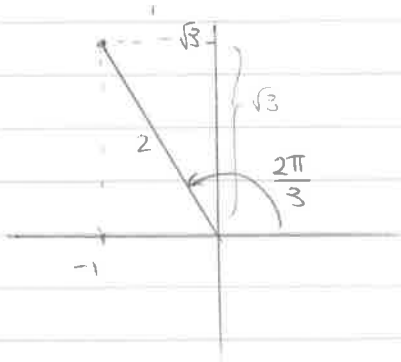
Q7. A

Working:

$$\begin{aligned} \text{Q1. } z\bar{z} &= (x+iy)(x-iy) \\ &= x^2 + y^2 \\ &= |z|^2 \end{aligned}$$

$$\therefore |z| = \sqrt{z\bar{z}} \quad \text{D}$$

Q2.



$$\begin{aligned} \arg(z) &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3} \end{aligned} \quad \text{C}$$

$$\begin{aligned} \text{Q3. } -(x^2 - 6x) &= 9 - (x^2 - 6x + 9) \\ &= 3^2 - (x-3)^2 \end{aligned}$$

$$\begin{aligned} \therefore \int \frac{1}{\sqrt{6x-x^2}} dx &= \int \frac{1}{\sqrt{3^2 - (x-3)^2}} dx \\ &= \sin^{-1}\left(\frac{x-3}{3}\right) + C \end{aligned} \quad C$$

$$\text{Q4. } \int \cos^3 x \sin x \, dx$$

$$= -\int \cos^3 x (-\sin x) \, dx$$

$$= -\frac{\cos^4 x}{4} + C \quad D$$

$$\text{Q5. } 2xy + x^2 \frac{dy}{dx} - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (2y - x^2) = 2xy$$

$$\therefore \frac{dy}{dx} = \frac{2xy}{2y - x^2} \quad D$$

$$\text{Q6. } \int \frac{dx}{e^x + 1} = \int \frac{e^x + 1 - e^x}{e^x + 1} dx$$

$$= \int \left(1 - \frac{e^x}{e^x + 1}\right) dx$$

$$= x - \ln(e^x + 1) + C \quad B$$

$$\begin{aligned} \text{Q7. } \frac{1}{z} &= 1 \operatorname{cis} 0 \div r \operatorname{cis} \theta \\ &= \frac{1}{r} \operatorname{cis}(-\theta) \end{aligned}$$

$$r \div \frac{2}{3} \quad \therefore \frac{1}{r} \div \frac{3}{2} \quad A$$

QUESTION 8

a) $w = -1 + i$ and $z = 4 + 3i$

i) $2w - \bar{z} = 2(-1 + i) - (4 - 3i)$
 $= -6 + 5i$ ✓

ii) $\operatorname{Im}\left(\frac{z}{i}\right) = \operatorname{Im}\left(\frac{4 + 3i}{i} \times \frac{-i}{-i}\right)$
 $= \operatorname{Im}(3 - 4i)$
 $= -4$ ✓

iii) $\frac{w}{z} = \frac{-1 + i}{4 + 3i} \times \frac{4 - 3i}{4 - 3i}$ ✓
 $= \frac{-4 + 3i + 4i + 3}{16 + 9}$

$$= -\frac{1}{25} + \frac{7}{25}i$$
 ✓

b) i) Let $(x + iy)^2 = 5 - 12i$

$$\left. \begin{array}{l} x^2 + y^2 = 5 \\ 2xy = -12 \\ xy = -6 \end{array} \right\}$$
 ✓

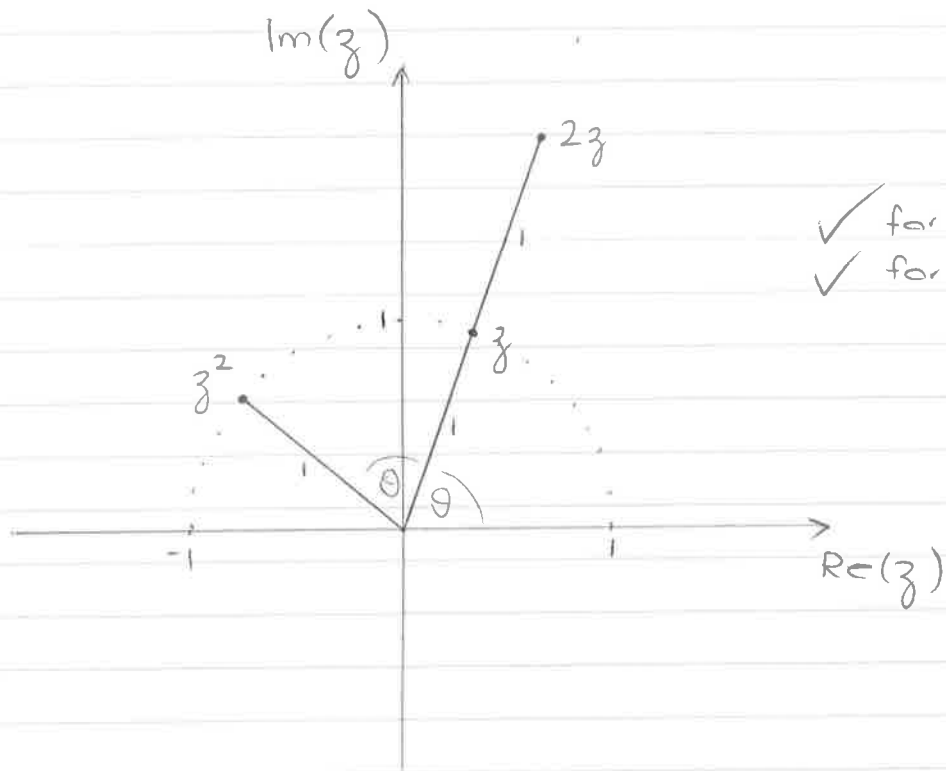
By inspection $x = 3$ or $x = -3$
 $y = -2$ $y = 2$

∴ the square roots are $3 - 2i$ or $-3 + 2i$ ✓

ii) $\Delta = (-(1 - 4i))^2 - 4 \times 1 \times -(5 - i)$
 $= 1 - 8i - 16 + 20 - 4i$
 $= 5 - 12i$ ✓

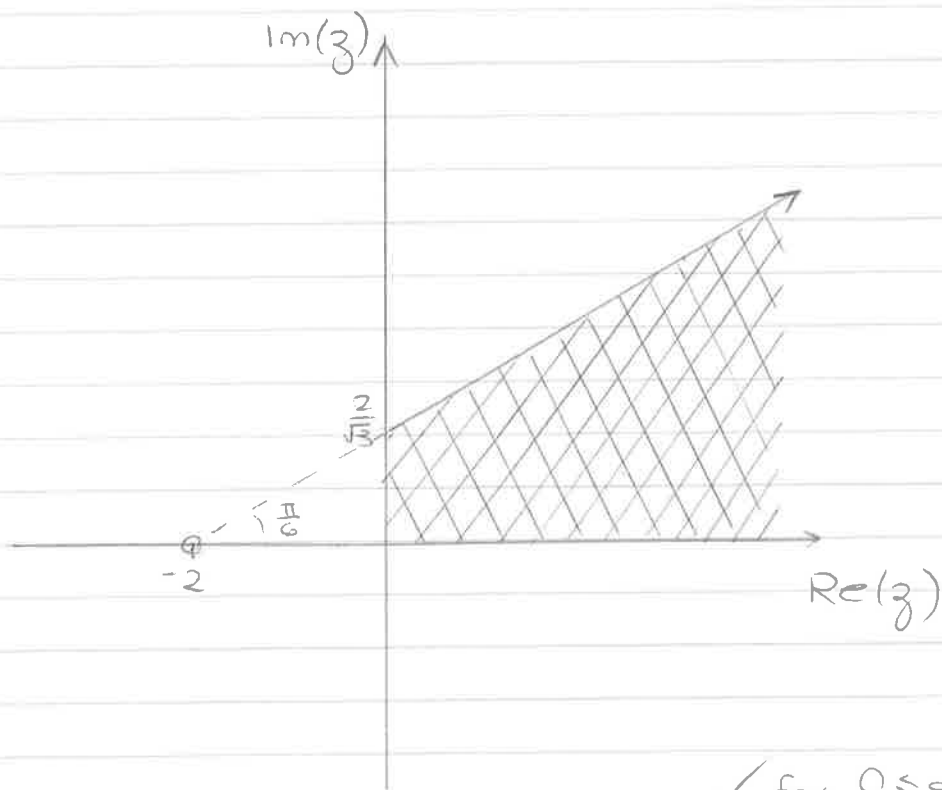
$$z = \frac{1 - 4i + 3 - 2i}{2} \quad \text{or} \quad \frac{1 - 4i - 3 + 2i}{2}$$
$$= 2 - 3i \quad -1 - i$$
 ✓

c)



✓ for two correct
✓ for all correct

d)



✓ for $0 \leq \arg(z+2) \leq \frac{\pi}{6}$
✓ for intersection
with $\text{Re}(z) \geq 0$

QUESTION 9:

$$\begin{aligned} \text{a) i) RHS} &= \frac{\sin(\alpha+\beta) - \sin(\alpha-\beta)}{2} \\ &= \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta - (\sin\alpha\cos\beta - \cos\alpha\sin\beta)}{2} \\ &= \frac{2\cos\alpha\sin\beta}{2} \quad \checkmark \text{ show that...} \\ &= \cos\alpha\sin\beta \\ &= \text{LHS} \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_0^\pi \cos 3x \sin 2x \, dx &= \int_0^\pi \frac{\sin(3x+2x) - \sin(3x-2x)}{2} \, dx \\ &= \frac{1}{2} \int_0^\pi (\sin 5x - \sin x) \, dx \quad \checkmark \\ &= \frac{1}{2} \left[-\frac{\cos 5x}{5} + \cos x \right]_0^\pi \\ &= \frac{1}{2} \left(-\frac{(-1)}{5} + (-1) - \left(-\frac{1}{5} + 1 \right) \right) \\ &= -\frac{4}{5} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) i) } \int \frac{6x^3 + 3x^2 - 2}{2x+1} \, dx &= \int \frac{3x^2(2x+1) - 2}{2x+1} \, dx \quad \checkmark \\ &= \int \left(3x^2 - \frac{2}{2x+1} \right) \, dx \\ &= x^3 - \ln|2x+1| + C \quad \checkmark \end{aligned}$$

$$b) \text{ ii) } \int \sin^3 x \cos^2 x \, dx$$

$$= \int \sin x (1 - \cos^2 x) \cos^2 x \, dx$$

$$= \int (\cos^2 x - \cos^4 x) \sin x \, dx \quad \checkmark$$

$$= -\int (u^2 - u^4) \, du$$

$$\text{Let } u = \cos x$$

$$du = -\sin x \, dx$$

$$= -\left(\frac{u^3}{3} - \frac{u^5}{5}\right) + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C \quad \checkmark$$

$$c) \int_1^e \frac{(\log_e x)^2}{x} \, dx$$

$$\text{Let } u = \log_e x$$

$$du = \frac{1}{x} \, dx \quad \checkmark$$

$$= \int_0^1 u^2 \, du$$

$$= \left[\frac{u^3}{3}\right]_0^1$$

$$= \frac{1}{3} \quad \checkmark$$

x	1	e
u	0	1

$$d) \int_0^{\frac{\pi}{3}} \frac{1}{2 - \cos x} \, dx$$

$$\text{Let } t = \tan \frac{x}{2}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{2 - \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} \, dt$$

$$x = 2 \tan^{-1} t$$

$$dx = \frac{2}{1+t^2} \, dt$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{2 + 2t^2 - 1 + t^2} \, dt$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{1+3t^2} \, dt \quad \checkmark$$

$$= \frac{2}{3} \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{\left(\frac{1}{\sqrt{3}}\right)^2 + t^2}$$

$$= \frac{2}{3} \left[\sqrt{3} + \tan^{-1}(\sqrt{3}t) \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= \frac{2\sqrt{3}}{3} \left(\tan^{-1} 1 - \tan^{-1} 0 \right) = \frac{2\sqrt{3}}{3} \times \frac{\pi}{4} = \frac{\pi\sqrt{3}}{6} \quad \checkmark$$

x	0	$\frac{\pi}{3}$
t	0	$\frac{1}{\sqrt{3}}$

QUESTION 10:

$$a) i) \frac{x}{(x^2+1)(x+1)(x-1)} \equiv \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$

$$x = A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x+1)(x-1)$$

$$\text{Sub } x=1: 1 = 4A$$

$$\therefore A = \frac{1}{4}$$

$$\text{Sub } x=-1: -1 = -4B$$

$$\therefore B = \frac{1}{4}$$

$$\text{Sub } x=0: 0 = \frac{1}{4} - \frac{1}{4} - D$$

$$\therefore D = 0$$

$$\text{Coeff of } x^3: 0 = \frac{1}{4} + \frac{1}{4} + C$$

$$\therefore C = -\frac{1}{2}$$

$$ii) \int \frac{x}{x^4-1} dx = \int \left(\frac{1}{4(x-1)} + \frac{1}{4(x+1)} - \frac{x}{2(x^2+1)} \right) dx$$

$$= \frac{1}{4} \int \left(\frac{1}{x-1} + \frac{1}{x+1} - \frac{2x}{x^2+1} \right) dx$$

$$= \frac{1}{4} \left(\ln|x-1| + \ln|x+1| - \ln(x^2+1) \right) + C$$

$$= \frac{1}{4} \ln \left| \frac{(x-1)(x+1)}{x^2+1} \right| + C$$

$$b) \vec{BD} = \vec{BC} + \vec{CD}$$

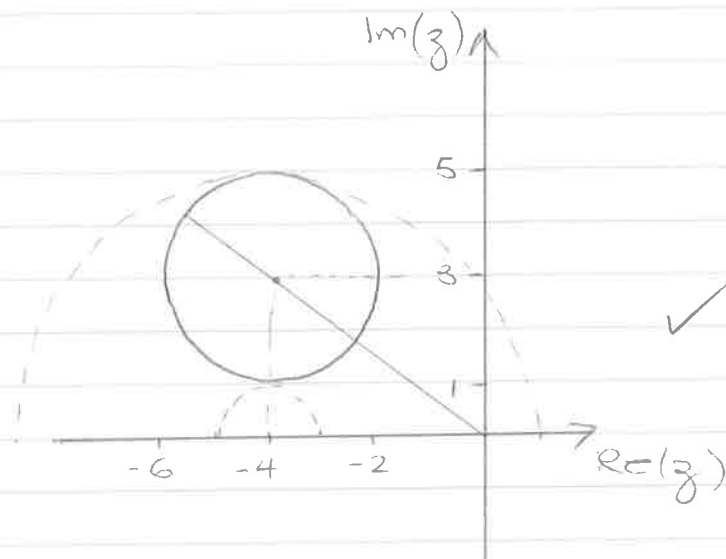
$$= \vec{BC} + \vec{BA}$$

$$\left(\vec{CD} = \vec{BA} \text{ since } ABCD \text{ is a parallelogram} \right)$$

$$= \vec{z}_3 - \vec{z}_2 + \vec{z}_1 - \vec{z}_2$$

$$= \vec{z}_1 - 2\vec{z}_2 + \vec{z}_3$$

$$c) |z - (-4 + 3i)| = 2$$



$$ii) |z|_{\max} = \sqrt{4^2 + 3^2} + 2 = 7$$

$$iii) |z - (-4)| = k$$

$$k = 1 \text{ or } 5$$

$$d) i) \text{ Let } u = \sqrt{x}$$

$$\therefore x = u^2$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$\begin{aligned} \int e^{\sqrt{x}} dx &= \int e^u \times 2u du \\ &= \int 2ue^u du \end{aligned}$$

$$d) \text{ ii) } \int_0^4 e^{\sqrt{x}} dx$$

$$u = \sqrt{x}$$

$$= \int_0^2 2u e^u du$$

x	0	4
u	0	2

$$= 2 \int_0^2 u \frac{d}{du} (e^u) du$$

$$= 2 \left\{ [u e^u]_0^2 - \int_0^2 1 \times e^u du \right\}$$

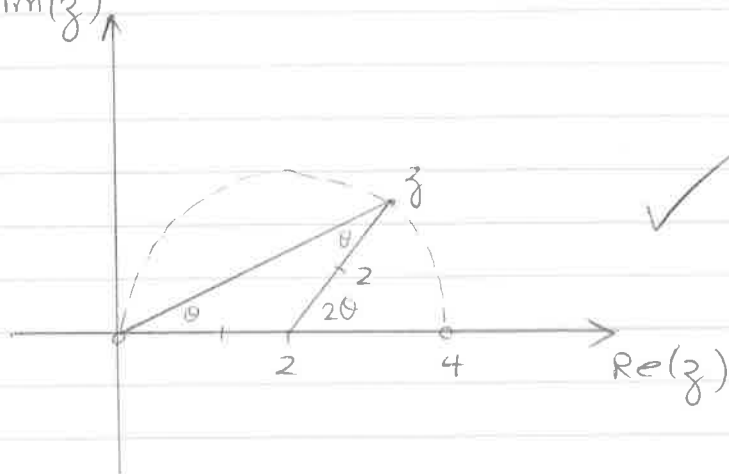
$$= 2 (2e^2 - 0 - [e^u]_0^2)$$

$$= 2 (2e^2 - (e^2 - e^0))$$

$$= 2 (e^2 + 1)$$

QUESTION 11:

a) i) $\text{Im}(z)$



$$\arg z = \theta$$

$$\arg(z-2) = 2\theta$$

$$= 2\arg z$$

angles at centre and circumference on the same arc.

or

external angle of isosceles triangle

$$\text{ii) } \arg(z^2 - 2z) = \arg(z(z-2))$$

$$= \arg z + \arg(z-2)$$

$$= \frac{\arg(z-2) + \arg(z-2)}{2}$$

$$= \frac{3\arg(z-2)}{2}$$

$$\text{b) i) } f(x) = \tan^{-1}\left(x + \frac{1}{x}\right)$$

$$f'(x) = \frac{1}{1 + \left(x + \frac{1}{x}\right)^2} \times \left(1 - \frac{1}{x^2}\right)$$

$$= 0 \text{ when } 1 - \frac{1}{x^2} = 0$$

$$\therefore x = \pm 1$$

$$\text{BUT } x > 0$$

$$\therefore x = 1$$

$$y = \tan^{-1} 2$$

$\therefore$ stationary point at $(1, \tan^{-1} 2)$

$$\begin{aligned} \text{ii)} \quad y &= \lim_{x \rightarrow \infty} \tan^{-1}\left(x + \frac{1}{x}\right) \\ &= \lim_{x \rightarrow \infty} \tan^{-1} x \\ &= \frac{\pi}{2} \end{aligned}$$

$\therefore$ horizontal asymptote at $y = \frac{\pi}{2}$. ✓

$$\begin{aligned} \text{iii)} \quad \lim_{x \rightarrow 0^+} \tan^{-1}\left(x + \frac{1}{x}\right) &= \lim_{x \rightarrow \infty} \tan^{-1}\left(x + \frac{1}{x}\right) \\ \therefore \lim_{x \rightarrow 0} f(x) &= \frac{\pi}{2} \end{aligned}$$

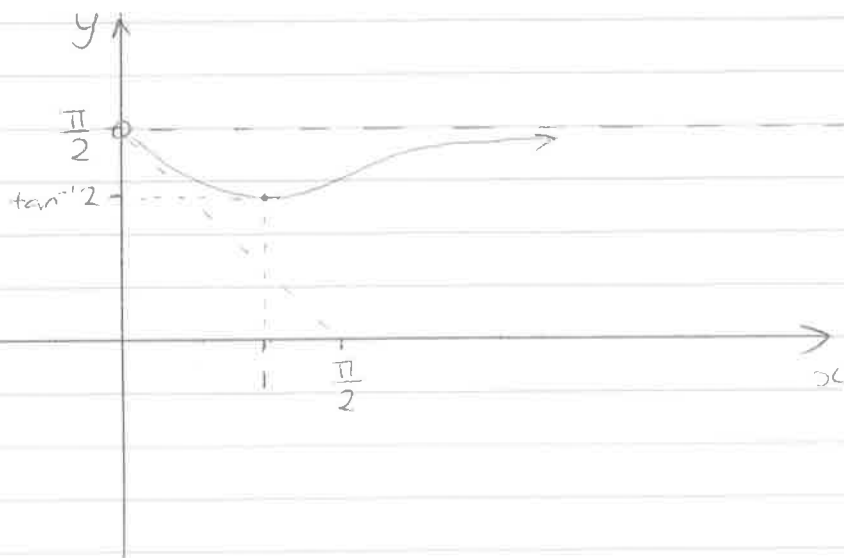
$$f'(x) = \frac{1}{1 + \left(x + \frac{1}{x}\right)^2} \times \left(1 - \frac{1}{x^2}\right)$$

$$= \frac{1}{1 + x^2 + 2 + \frac{1}{x^2}} \times \frac{x^2 - 1}{x^2}$$

$$= \frac{x^2 - 1}{x^4 + 3x^2 + 1} \quad x \neq 0$$

$$\begin{aligned} \therefore \lim_{x \rightarrow 0} f'(x) &= \frac{-1}{1} \\ &= -1 \end{aligned}$$

iv)



$$c) i) x^2 + y^2 + (1-2i)(x+iy) + (1+2i)(x-iy) \leq 3$$

$$x^2 + y^2 + x + iy - 2ix + 2y + x - iy + 2ix + 2y \leq 3$$

$$x^2 + 2x + 1 + y^2 + 4y + 4 \leq 3 + 1 + 4$$

$$(x+1)^2 + (y+2)^2 \leq 8$$



OR //

$$(z + (1+2i))(\overline{z + (1+2i)}) \leq 3 + |1+2i|^2$$

$$|z + 1 + 2i|^2 \leq 8$$

$$|z - (-1-2i)| \leq 2\sqrt{2}$$

OR //

$$z\bar{z} + (1-2i)z + \overline{(1-2i)z} \leq 3$$

$$z\bar{z} + 2\operatorname{Re}((1-2i)z) \leq 3$$

$$x^2 + y^2 + 2\operatorname{Re}((1-2i)(x+iy)) \leq 3$$

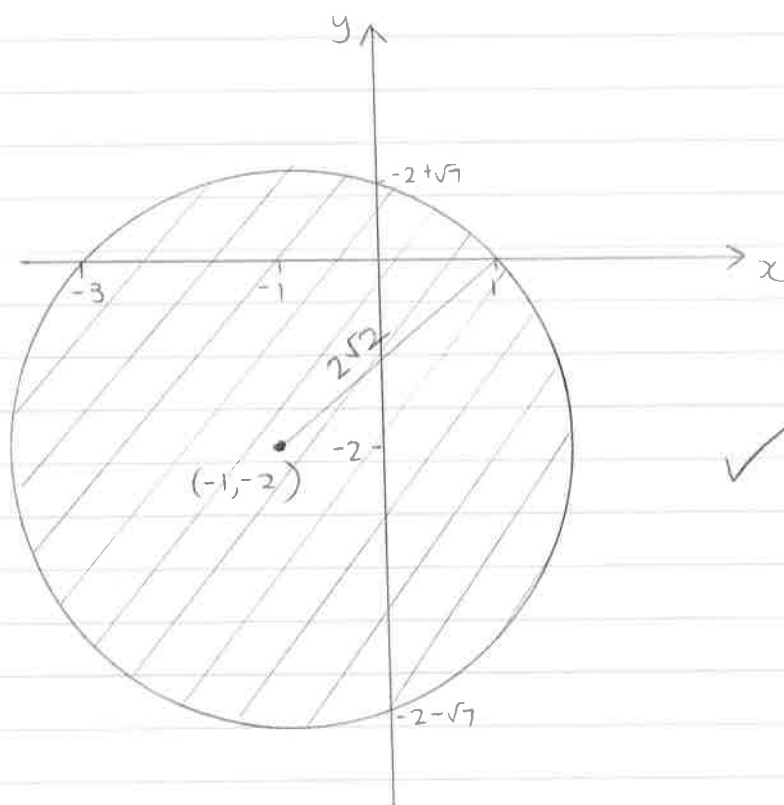
$$x^2 + y^2 + 2(x + 2y) \leq 3$$

$$x^2 + 2x + 1 + y^2 + 4y + 4 \leq 3 + 1 + 4$$

$$(x+1)^2 + (y+2)^2 \leq 8$$



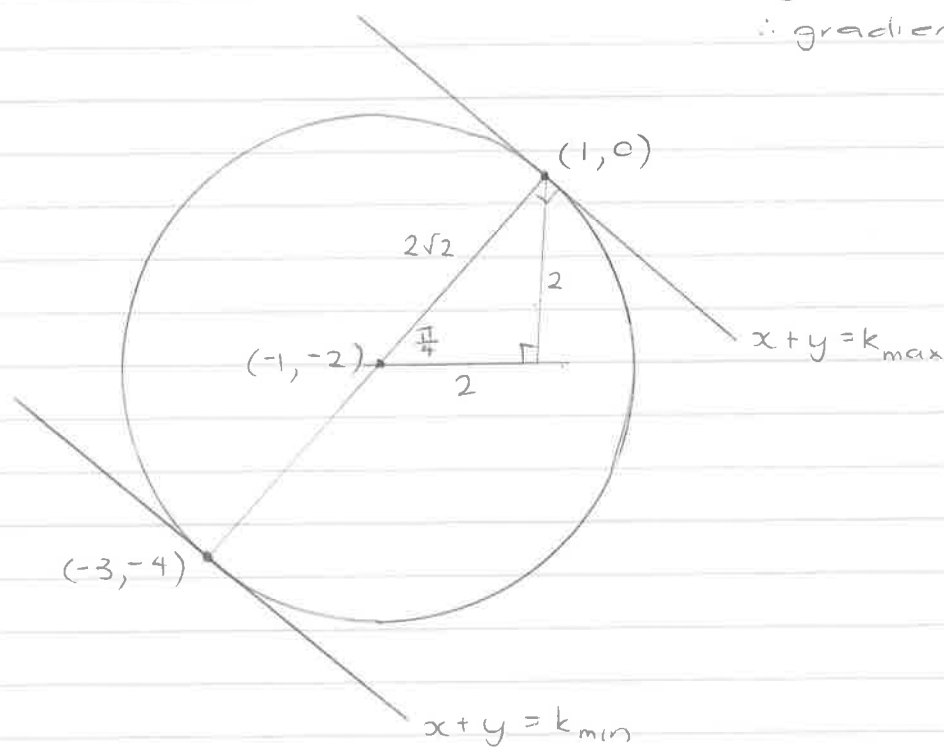
c) i) cont...



✓ (intercept values not req'd for mark)

ii) the max & min values will occur when the line is tangent to the circle.

gradient of $x+y=k$ is -1
 $\therefore$ gradient of perpendicular is 1 .



$(1, 0)$ satisfies equation $\therefore (x+y)_{\max} = 1+0 = 1$

$(-3, -4)$ satisfies equation $\therefore (x+y)_{\min} = -3-4 = -7$

✓